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SINGLE-INPUT, MULTI-OUTPUT DIRECT-CURRENT-TO-DIRECT CURRENT CONVERTERS AND ASSOCIATED METHODS

Abstract

A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes a (i) power conversion stage electrically coupled between an input power node and a switching node and (ii) a plurality of output switching blocks. Each output switching block is electrically coupled between the switching node and a respective output power node. Each output switching block includes a respective input N-type metal oxide semiconductor field effect transistor (input NMOS FET) and a respective output N-type metal oxide semiconductor field effect transistor (output NMOS FET). The input NMOS FET is oriented in each output switching block such that a cathode of a body diode of the input NMOS FET is electrically coupled to the switching node. The output NMOS FET is oriented in each output switching block such that a cathode of a body diode of the output NMOS FET is electrically coupled to the output power node of the output switching block.

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Background/Summary

BACKGROUND

[0001] Many electrical devices require multiple power supply rails. For example, an electrical device may require two or more power supply rails to power circuitry having different power supply voltage requirements. As another example, an electrical device may require two or more power supply rails to enable two or more systems within the electrical device to be independently powered on and powered off.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a schematic diagram of a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, according to an embodiment.

[0003] FIG. 2 is a schematic diagram of a first output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0004] FIG. 3 is a schematic diagram of a second output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0005] FIG. 4 is a schematic diagram of an Nth output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0006] FIG. 5 is a schematic diagram of one embodiment of the first output switching block of FIG. 2.

[0007] FIG. 6 is a schematic diagram of one embodiment of the second output switching block of FIG. 3.

[0008] FIG. 7 is a schematic diagram of one embodiment of the Nth output switching block of FIG. 4.

[0009] FIG. 8 includes three graphs collectively illustrating example control signals for a power conversion stage of the FIG. 1 SIMO DC-to-DC converter when the power conversion stage operates as a boost converter.

[0010] FIG. 9 includes three graphs collectively illustrating example control signals for a power conversion stage of the FIG. 1 SIMO DC-to-DC converter when the power conversion stage operates as a buck converter.

[0011] FIG. 10 is a state diagram illustrating example operating states of the FIG. 1 SIMO DC-to-DC converter.

[0012] FIG. 11 includes five graphs collectively illustrating an example of an output switching block transitioning from a first off-state to an on-state.

[0013] FIG. 12 includes five graphs collectively illustrating an example of an output switching block transitioning from the on-state to the first off-state.

[0014] FIG. 13 includes five graphs collectively illustrating an example of an output switching block transitioning from a second off-state to the on-state.

[0015] FIG. 14 includes five graphs collectively illustrating an example of an output switching block transitioning from the on-state to the second off-state.

[0016] FIG. 15 is a schematic diagram of an alternate embodiment of the FIG. 5 output switching block that further includes elements to perform current sensing.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0017] A plurality of power supply rails in an electrical device can be provided by providing a respective power converter, such as a respective direct-current-to-direct-current (DC-to-DC) converter, for each power supply rail. However, providing a respective power converter for each

power supply rail may be expensive and require significant space. Therefore, it is frequently desirable to provide multiple power supply rails from a single power converter. For example, a single-input, multi-output (SIMO) DC-to-DC converter may be used to provide multiple output power nodes from a single input power node. The output power nodes, though, require output switching circuitry with bidirectional current blocking capability to ensure that each output power node is electrically isolated from each other output power node, as well as to enable control of which output power node is powered by the SIMO DC-to-DC converter at a given time.

[0018] It is frequently desirable to implement a switch, such as in output switching circuitry of a SIMO DC-to-DC converter, using an enhancement mode, N-type metal oxide semiconductor field effect transistor (NMOS FET) because an NMOS FET generally has lower on-resistance than an enhancement mode, P-type metal oxide semiconductor field effect transistor (PMOS FET) of similar size. Additionally, NMOS FET fabrication is highly developed in modern semiconductor fabrication processes. However, an NMOS FET includes a body diode between its source and drain which generally prevents the NMOS FET from blocking bidirectional current. While this issue can be overcome by managing the body diode through body switching, body switching is risky and non-robust because the body is floating for a short time during switching, which impairs breakdown voltage rating of the NMOS FET. Additionally, body switching necessitates complex gate drive schemes, and there may be limited support for body switching in a semiconductor fabrication process.

[0019] As such, it is generally desirable to electrically couple two NMOS FETs in series with opposing body diode orientations to enable bidirectional current blocking, instead of implementing body switching. However, gate drive of two series-connected NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter is non-trivial. For example, one conventional approach for driving gates of NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter requires a respective bootstrap driving domain referred to the switching node and a respective power supply rail for each output power node referred to the output power node itself, which may be costly, space consuming, and quiescent current demanding. As another example, another conventional approach for driving gates of NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter relies on a single power supply rail to power high-side NMOS FETs across multiple output power nodes, which may limit operating voltage range and/or be unable to achieve high performance.

[0020] Disclosed herein are new SIMO DC-to-DC converters which at least partially overcome the above discussed drawbacks. The new SIMO DC-to-DC converters include a respective output switching block for each output power node where each output switching block requires only a single respective power supply rail, thereby promoting low SIMO DC-to-DC converter cost and small SIMO DC-to-DC converter size. Additionally, NMOS FETs are oriented in the output switching blocks of the new SIMO DC-to-DC converters in a manner that promotes robust and simple gate drive of the NMOS FETs with a single power supply rail, thereby helping achieve high performance, such as low on-resistance. Furthermore, the new SIMO DC-to-DC converters do not require field effect transistor body switching, thereby further promoting robust operation. Moreover, particular embodiments of the new SIMO DC-to-DC converters can achieve high performance operation over a wider voltage range than can be realized by typical conventional SIMO DC-to-DC converters. For example, certain embodiments are easily scalable to operate over different voltage ranges simply by changing the drain-to-source and drain-to-gate voltage ratings of constituent NMOS FETs, which are frequently available in a wide range of ratings in typical semiconductor fabrication processes.

[0021] FIG. 1 is a schematic diagram of a SIMO DC-to-DC converter **100**, which is one embodiment of the new SIMO DC-to-DC converters disclosed herein. The SIMO DC-to-DC converter **100** includes a power conversion stage **102**, N output switching blocks **104**, and a controller **106**, where N is an integer greater than or equal to two. In this document, specific

instances of an item may be referred to by use of a numeral in parentheses (e.g. the output switching block **104(1)**) while numerals without parentheses refer to any such item (e.g. the output switching blocks **104**). Although FIG. **1** illustrates N being at least three, it is understood that N could be equal to two. The power conversion stage **102** is electrically coupled between an input power node **108** and a switching node **110**, and each output switching block **104** is electrically coupled between the switching node **110** and a respective output power node **112**. Each output power node implements, for example, a respective power supply rail that electrically powers one or more loads (not shown). The SIMO DC-to-DC converter **100** can, and typically will, include additional components, such as one or more capacitors (not shown), electrostatic discharge (ESD) protection circuitry (not shown), etc.

[0022] The input power node **108** has an input voltage $V_{sub.in}$ with respect to a reference node **114**, and each output power node **112** has a respective output voltage $V_{sub.out}$ with respect to the reference node **114**. While not required, in some embodiments, nominal magnitude of the output voltage $V_{sub.out}$ varies among two or more of the output power nodes **112**, such as to enable the SIMO DC-to-DC converter **100** to power two or more loads requiring different power supply voltage magnitude. The reference node **114** is depicted as being a ground node, such as an earth ground node or a chassis ground node. It is understood, though, that the reference node **114** need not be a ground node, and the reference node **114** accordingly could be at a different electrical potential than an earth ground or a chassis ground. An input current $I_{sub.in}$ flows into the SIMO DC-to-DC converter **100** from the input power node **108**, and a respective output current I_{out} flows out of each output power node **112** to a load (not shown) electrically coupled to the output power node **112**. A respective polarity of the input current $I_{sub.in}$ and each output current I_{out} could be either positive or negative, depending on the operating conditions of the SIMO DC-to-DC converter **100**.

[0023] The power conversion stage **102** includes a power conversion switching device **116**, a power conversion switching device **118**, a power conversion switching device **120**, and an inductor **122**. Each of the aforementioned power conversion switching devices includes, for example, one or more transistors and associated driver circuitry. The power conversion switching device **116** is electrically coupled between the input power node **108** and a connection node **124**, and the power conversion switching device **118** is electrically connected between the connection node **124** and the reference node **114**. The inductor **122** is electrically coupled between the connection node **124** and the switching node **110**, and the power conversion switching device **120** is electrically coupled between the switching node **110** and the reference node **114**. The power conversion switching device **116** is controlled by a signal $pc1$ generated by the controller **106**, the power conversion switching device **118** is controlled by a signal $pc2$ generated by the controller **106**, and the power conversion switching device **120** is controlled by a signal $pc3$ generated by the controller **106**. As discussed below, the controller **106** generates the control signal $pc1$, the control signal $pc2$, and/or the control signal $pc3$ to cause the power conversion stage **102** to provide a regulated voltage $V_{sub.s}$ with respect to the reference node **114** at the switching node **110**.

[0024] As discussed below, each output switching block **104** is configured to connect and disconnect its respective output power node **112** from the switching node **110** under the command of the controller **106**. Additionally, each output switching block **104** is capable of bidirectional current blocking and thereby electrically isolating its respective output power node **112** from each other output power node **112**. Each output switching block **104** is at least partially electrically powered by an auxiliary power supply rail **126** having a voltage V_{aux} . Each output switching block **104** is also electrically coupled to the reference node **114**. In some embodiments, each output switching block **104** is electrically powered from the input power node **108** instead of from the auxiliary power supply rail **126**, and the auxiliary power supply rail **126** is therefore optionally omitted. Details of the output switching blocks **104** are not shown in FIG. **1**, but FIGS. **2**, **3**, and **4**, illustrate details of the output switching blocks **104(1)**, **104(2)**, and **104(N)**, respectively. It is

understood that other instances of the output switching blocks **104** are embodied in a manner analogous to the output switching blocks **104(1)**, **104(2)**, and **104(N)**.

[0025] Referring to FIGS. **2-4**, each output switching block **104** includes an input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) **202**, an output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) **204**, input switch driver circuitry **206**, output switch driver circuitry **208**, bootstrap circuitry **210**, and a reference switching device **212**. Each input NMOS FET **202** and each output NMOS FET **204** has a respective gate (G), a respective drain (D), and a respective source(S). Additionally, each input NMOS FET **202** and each output NMOS FET **204** includes a respective body diode **216** and **218** with an anode (A) and a cathode (K). In each output switching block **104**, the input NMOS FET **202** is electrically coupled between the switching node **110** and an internal node **214** of the output switching block **104** with an orientation such that (i) the drain of the input NMOS FET **202**, as well as the cathode of the body diode **216**, are electrically coupled to the switching node **110**, and (ii) the source of the input NMOS FET **202**, as well as the anode of the body diode **216**, are electrically coupled to the internal node **214** of the output switching block **104**. Additionally, in each output switching block **104**, the output NMOS FET **204** is electrically coupled between the internal node **214** of the output switching block **104** and the output power node **112** of the output switching block **104** with an orientation such that (i) the drain of the output NMOS FET **204**, as well as the cathode of the body diode **218**, are electrically coupled to the output power node **112**, and (ii) the source of the output NMOS FET **204**, as well as the anode of the body diode **218**, are electrically coupled to the internal node **214** of the output switching block **104**.

[0026] The bootstrap circuitry **210** of each output switching block **104** generates a respective bootstrap power supply rail **220** of the output switching block **104** having a voltage $V_{sub.b}$ that is referenced to the internal node **214** of the output switching block **104**. The input switch driver circuitry **206** of each output switching block **104** drives the gate of the input NMOS FET **202** of the output switching block **104** in response to a respective control signal s_{bin} generated by the controller **106** (FIG. **1**). Additionally, the output switch driver circuitry **208** of each output switching block **104** drives the gate of the output NMOS FET **204** of the output switching block **104** in response to a respective control signal s_{bout} generated by the controller **106** (FIG. **1**). In each output switching block **104**, the input switch driver circuitry **206** and the output switch driver circuitry **208** are (i) powered from the bootstrap power supply rail **220** of the output switching block **104** and (ii) electrically reference to the internal node **214** of the output switching block **104** (instead of to the reference node **114**). As such, in each output switching block **104**, the input switch driver circuitry **206** drives the gate of the input NMOS FET **202** between the voltage $V_{sub.b}$ of the bootstrap power supply rail **220** and the voltage V_{int} of the internal node **214**, and the output switch driver circuitry **208** drives the gate of the output NMOS FET **204** between the voltage $V_{sub.b}$ of the bootstrap power supply rail **220** and the voltage V_{int} of the internal node **214**. The reference switching device **212** of each output switching block **104** is electrically coupled between the internal node **214** of the output switching block **104** and the reference node **114**.

[0027] The above-described orientations of the input NMOS FETs **202** and the output NMOS FETs **204** are important. In particular, not only do the orientations enable bidirectional current blocking by preventing current flow through an output switching block **104** solely via the body diodes **216** and **218**, but the orientations also facilitate gate drive of the NMOS FETs. Specifically, the above-describe orientations of the input NMOS FET **202** and the output NMOS FET **204** in an output switching block **104** result in the source of each of the two FETs to be connected to a common node, i.e., the internal node **214** of the output switching block **104**, which enables the respective gates of the two FETs to receive a common gate-to-source voltage when being driven by a common power supply rail, i.e. bootstrap power supply rail **220**, which facilitates robust driving of the two FETs from the common power supply rail.

[0028] FIGS. **5-7** illustrate one possible embodiment of bootstrap circuitry **210** and reference

switching device **212** of output switching blocks **104**. Specifically, FIG. 5 is a schematic diagram of an output switching block **504(1)**, which is an embodiment of output switching block **104(1)** (FIG. 2), FIG. 6 is a schematic diagram of an output switching block **504(2)**, which is an embodiment of output switching block **104(2)** (FIG. 3), and FIG. 7 is a schematic diagram of an output switching block **504(N)**, which is an embodiment of output switching block **104(N)** (FIG. 4). In the output switching blocks of FIGS. 5-7, bootstrap circuitry **210** is embodied by bootstrap circuitry **510**, and the reference switching device **212** is embodied by a reference enhancement mode, N-type metal oxide semiconductor field effect transistor (reference NMOS FET) **512**.

[0029] In each output switching block **504**, the bootstrap circuitry **510** includes a bootstrap capacitor **522** and a switching device embodied by a PMOS FET **524** and bootstrap switching device driver circuitry **526**. The bootstrap capacitor **522** of each output switching block **504** is electrically coupled between the internal node **214** of the output switching block **504** and the bootstrap power supply rail **220** of the output switching block **504**. Additionally, in each output switching block **504**, the PMOS FET **524** is electrically coupled between the auxiliary power supply rail **126** and the bootstrap power supply rail **220** of the output switching block **504**, and the bootstrap switching device driver circuitry **526** drives a gate of the PMOS FET **524** in response to a signal *bc* generated by the controller **106** (FIG. 1). The bootstrap switching device driver circuitry **526** of each output switching block **504** is (i) powered from the bootstrap power supply rail **220** of the output switching block **504** and (ii) is electrically referenced to the internal node **214** of the output switching block **504**. As such, all switching device driver circuitry of a given output switching block **504** operates in a common power supply domain, i.e., between the bootstrap power supply rail **220** of the output switching block **504** and the internal node **214** of the output switching block **504**.

[0030] A drain of the reference NMOS FET **512** in each output switching block **504** is electrically coupled to the internal node **214** of the output switching block **504**, and the source of the reference NMOS FET **512** in each output switching block **504** is electrically coupled to the reference node **114**. The gate of each reference NMOS FET **512** is driven by a respective control signal *rc* generated by the controller **106** (FIG. 1).

[0031] Referring again to FIG. 1, the controller **106** is configured to generate the control signals for the power conversion stage **102** as well as the control signals for each output switching block **104**. Connections between the controller **106** and the power conversion stage **102**, as well as connections between the controller **106** and the output switching blocks **104**, are not shown for illustrative clarity. The controller **106** is implemented, for example, by analog and/or electronic circuitry. In some embodiments, the controller **106** is at least partially implemented by a processor (not shown) executing instructions in the form of software and/or firmware stored in a memory (not shown). Although the controller **106** is depicted as a discrete element for illustrative simplicity, the controller **106** could be partially or fully integrated with one or more other elements of the SIMO DC-to-DC converter **100**. For example, some subsystems of the controller **106** could be incorporated in the power conversion stage **102** and/or in one or more of the output switching blocks **104**. Additionally, the controller **106** may include multiple constituent elements that need not be co-packaged or even disposed at a common location.

[0032] As mentioned above, the controller **106** is configured to generate the control signal *pc1*, the control signal *pc2*, and/or the control signal *pc3* to cause the power conversion stage **102** to provide a regulated voltage $V_{sub.s}$ with respect to the reference node **114** at the switching node **110**. The power conversion stage **102** has a buck and boost topology, and the magnitude of the voltage $V_{sub.s}$ may therefore be greater than the magnitude of the voltage $V_{sub.in}$, less than magnitude of the voltage $V_{sub.in}$, or equal to magnitude of the voltage $V_{sub.in}$. For example, the controller **106** may cause the power conversion stage **102** to operate as a boost converter such that the magnitude of voltage $V_{sub.s}$ is greater than the magnitude of the voltage $V_{sub.in}$ by (i) generating the control signal *pc1* to cause the power conversion switching device **116** to continuously operate in its on-

state, (ii) generating the control signal pc2 to cause the power conversion switching device 118 to continuously operate in its off-state, and (iii) generating the control signal pc3 to modulate a duty cycle of the power conversion switching device 120 to achieve a desired magnitude of the voltage $V_{sub.s}$. The duty cycle of a switching device is a portion of each switching cycle of the switching device that the switching device operates in its on-state. For example, if the power conversion switching device 120 operates in its on-state for 40 percent of its switching period, the power conversion switching device 120 has a duty cycle of 40 percent. In this document, a transistor or other switching device is in its on-state when the transistor or other switching device is being controlled to operate in its conductive state. Conversely, a transistor or other switching device is in its off-state when the transistor or other switching device is being controlled to operate in its non-conductive state.

[0033] FIG. 8 includes graphs 800, 802, and 804 collectively illustrating one example of the control signals pc1, pc2, and pc3 generated by the controller 106 when the controller 106 is causing the power conversion stage 102 to act as a boost converter such that the magnitude of the voltage $V_{sub.s}$ is greater than the magnitude of the voltage $V_{sub.in}$. Graph 800 is of the control signal pc1 versus time, graph 802 is of the control signal pc2 versus time, and graph 804 is of the control signal pc3 versus time. Graphs 800, 802, and 804 have a common time base. FIG. 8 assumes that each of the power conversion switching device 116, the power conversion switching device 118, and the power conversion switching device 120 (i) operates in its on-state when its respective control signal is logic high and (ii) operates in its off-state when its respective control signal is logic low. However, the power conversion switching devices of power conversion stage 102 could be configured to respond to control signals in other manners without departing from the scope hereof. In the FIG. 8 example, the control signal pc1 is continuously logic high such that the power conversion switching device 116 continuously operates in its on-state, and the control signal pc2 is continuously logic low such that the power conversion switching device 118 continuously operates in its off-state. Additionally, the controller 106 generates the control signal pc3 in the FIG. 8 example such that (i) the control signal pc3 has a switching period $T_{sub.a}$, (ii) the control signal pc3 is logic high for a duration of $t_{sub.on_a}$ of each switching period $T_{sub.a}$, and (iii) the duration of $t_{sub.on_a}$ is 40 percent of the switching period $T_{sub.a}$. As such, the power conversion switching device 120 has a duty cycle of 40 percent in the FIG. 8 example.

[0034] Referring again to FIG. 1, controller 106 may cause the power conversion stage 102 to operate as a buck converter such that the magnitude of the voltage $V_{sub.s}$ is less than or equal to the magnitude of the voltage $V_{sub.in}$ by (i) generating the control signal pc1 to modulate a duty cycle of the power conversion switching device 116 to achieve a desired magnitude of voltage $V_{sub.s}$, (ii) generating the control signal pc2 to cause the power conversion switching device 118 to perform a freewheeling function, i.e., to provide a path for the current $I_{sub.L}$ flowing through the inductor 122 when the power conversion switching device 116 is in its off-state, and (iii) generating the control signal pc3 to cause the power conversion switching device 120 to continuously operate in its off-state. For example, FIG. 9 includes graphs 900, 902, and 904 collectively illustrating one example of control signals pc1, pc2, and pc3 generated by the controller 106 when the controller 106 is causing the power conversion stage 102 to act as a buck converter such that the magnitude of the voltage $V_{sub.s}$ is less than the magnitude of the voltage $V_{sub.in}$. Graph 900 is of the control signal pc1 versus time, graph 902 is of the control signal pc2 versus time, and graph 904 is of the control signal pc3 versus time. The graphs 900, 902, and 904 have a common time base. FIG. 9 assumes that each of the power conversion switching device 116, the power conversion switching device 118, and the power conversion switching device 120 (i) operates in its on-state when its respective control signal is logic high and (ii) operates in its off-state when its respective control signal is logic low. FIG. 9 also assumes that the current $I_{sub.L}$ through the inductor 122 is continuous.

[0035] In the FIG. 9 example, the controller 106 generates the control signal pc1 in the FIG. 9

example such that (i) the control signal pc1 has a switching period T_b , (ii) the control signal pc1 is logic high for a duration of $t_{sub.on_b}$ of each switching period T_b , and (iii) the duration of $t_{sub.on_b}$ is 60 percent of the switching period T_b . As such, the power conversion switching device **116** has a duty cycle of 60 percent in the FIG. 9 example. The control signal pc2 is complementary to the control signal pc1, and the control signal pc3 is continuously logic low, in the FIG. 9 example. Certain embodiments of the controller **106** are configured to insert deadtime between de-assertion of control signal pc1 and assertion of control signal pc2, and vice versa, to prevent shoot-through, i.e., simultaneous conduction of each of the power conversion switching device **116** and the power conversion switching device **118**.

[0036] Referring again to FIG. 1, the controller **106** may vary operation of the power conversion stage **102** according to which output power node **112** is being powered by the SIMO DC-to-DC converter **100**. For example, assume that (i) the SIMO DC-to-DC converter **100** is powering output power node **112(1)** and (ii) a load powered from the output power node **112(1)** requires a power supply voltage having a magnitude that is greater than the magnitude of the input voltage $V_{sub.in}$. In this scenario, the controller **106** may cause the power conversion stage **102** to operate as a boost converter such that the magnitude of the voltage $V_{sub.s}$ is equal to the magnitude of the voltage required by the load powered from the output power node **112(1)**. Now assume that (i) the SIMO DC-to-DC converter **100** changes its operating state so that the SIMO DC-to-DC converter **100** is powering the output power node **112(2)** and (ii) a load powered from the output power node **112(1)** requires a power supply voltage having a magnitude that is smaller than the magnitude of the input voltage $V_{sub.in}$. The controller **106** may accordingly cause the power conversion stage **102** to change from operating as a boost converter to operating as a buck converter such that the magnitude of the voltage $V_{sub.s}$ is equal to the magnitude of the voltage required by the load powered from the output power node **112(2)**. In some embodiments, the SIMO DC-to-DC converter **100** changes its active output power node **112** at a high rate, and the controller **106** may therefore cause the power conversion stage **102** to change its operating state at a corresponding high rate.

[0037] The controller **106** is configured to generate the control signals to control the output switching blocks **104** such that only one output power node **112** is powered by the SIMO DC-to-DC converter **100** at a given time, although some embodiments of the SIMO DC-to-DC converter **100** also support an alternate operating mode where two or more of the output power nodes **112** having a common voltage $V_{sub.out}$ may be simultaneously powered. The controller **106** is also configured to control the output switching blocks **104** so that at least one output switching block **104** provides a path for the current $I_{sub.L}$ flowing through the inductor **122** when (i) a magnitude of the current $I_{sub.L}$ is greater than zero and (ii) the power conversion switching device **120** is in its off-state. An output switching block **104** may provide a path for the current $I_{sub.L}$ by providing a path for the current $I_{sub.L}$ to flow to a load connected to the respective output power node **112** of the output switching block **104** via the input NMOS FET **202** and output NMOS FET **204** of the output switching block **104**. Alternately, an output switching block **104** may provide a path for the current $I_{sub.L}$ to flow to the reference node **114** via the reference switching device **212** of the output switching block **104**. Certain embodiments of the controller **106** are configured so that either the power conversion switching device **120** or at least one instance of an input NMOS FET **202** is in its on-state at any given time to ensure that there is path for the current $I_{sub.L}$, irrespective of direction of flow of the current.

[0038] Additionally, in particular embodiments, the controller **106** is further configured such that an output NMOS FET **204** of a given output switching block **104** may operate in its on-state only if the power conversion switching device **120** is in its off-state and all other output NMOS FETs **204** are in their respective off-states. Furthermore, in some embodiments, the controller **106** is additionally configured such that (i) the power conversion switching device **120** may operate in its on-state only if all output NMOS FETs **204** are in their respective off-states and (ii) the power

conversion switching device **120** may switch from its on-state to its off-state only once the PMOS FET **524** and the reference NMOS FET **512** of an output switching block **504** most-recently powering its respective output power node **112** have transitioned from their respective on-states to their respective off-states. Moreover, in certain embodiments, the controller **106** is further configured such to cause each of the PMOS FET **524** and the reference NMOS FET **512** of a given output switching block **504** that is powering its respective output power node **112** to remain in their respective on-states for a time window enclosed within the turn-on time of the power conversion switching device **120** in case the related output switching block **504** has its input NMOS FET **202** turned on and its output NMOS FET **204** turned off, or continuously as long as the related output switching block **504** has its input NMOS FET **202** and its output NMOS FET **204** both turned off. [0039] FIG. **10** is a state diagram **1000** illustrating several possible operating states of each output switching block **104**, although it is understood that the output switching blocks **104** could operate in other operating states without departing from the scope hereof. FIG. **10** assumes that the output switching blocks **104** are embodied as output switching blocks **504** and that (i) each switching device of the SIMO DC-to-DC converter **100** operates in its on-state with its respective control signal is logic high and (ii) each switching device of the SIMO DC-to-DC converter **100** operates in its off-state with its respective control signal is logic low. State diagram **1000** illustrates three possible operating states of an output switching block **504**, i.e., an on-state **1002**, a first off-state **1004**, and a second off-state **1006**. The on-state **1002** of an output switching block **504** is characterized by the output switching block **504** electrically coupling its respective output power node **112** to the switching node **110**, such that the respective output power node **112** is powered by the SIMO DC-to-DC converter **100**. Accordingly, the control signal sbin and the control signal sbout of the output switching block **504** are logic high in the on-state **1002**, to cause each of the input NMOS FET **202** and the output NMOS FET **204** of the output switching block **504** to operate in its on-state. Control signal rc of the output switching block **504** is logic low in the on-state **1002** to cause the reference NMOS FET **512** of the output switching block **504** to operate in its off-state. Additionally, the control signal bc of the output switching block **504** is logic low to cause the PMOS transistor **524** of the output switching block **504** to operate in its off-state, such that the bootstrap power supply rail **220** of the output switching block **504** is isolated from auxiliary power supply rail **126**.

[0040] The first off-state **1004** of an output switching block **504** is characterized by the output switching block **504** isolating its respective output power node **112** from switching node **110**, as well as by the output switching block **504** isolating its respective output power node **112** from the other output power nodes **112**. Therefore, the respective output power node **112** of the output switching block **504** is not powered by the SIMO DC-to-DC converter **100** when the output switching block **504** is in the first off-state **1004**. Accordingly, the control signal sbin and the control signal sbout of the output switching block **504** are logic low in the first off-state **1004**, to cause each of the input NMOS FET **202** and the output NMOS FET **204** of the output switching block **504** to operate in its off-state. The control signal rc of the output switching block **504** is logic high in the first off-state **1004** to cause the reference NMOS FET **512** of the output switching block **504** to operate in its on-state. Additionally, the control signal bc of the output switching block **504** is logic high in the first off-state **1004** to cause the PMOS transistor **524** of the output switching block **504** to operate in its on-state. Consequently, the bootstrap capacitor **522** of the output switching block **504** charges from the auxiliary power supply rail **126** and through the PMOS FET **524** of the output switching block **504**, in the first off-state **1004** of the output switching block **504**. [0041] The second off-state **1006** of an output switching block **504** is characterized by the SIMO DC-to-DC converter **100** not powering the respective output power node **112** of the output switching block **504**. Accordingly, control signal sbout of the output switching block **504** is logic low in the second off-state **1006** to cause the output NMOS FET **204** of the output switching block **504** to operate in its off-state. However, control signal sbin of the output switching block **504** is

logic high in the second off-state **1006** to cause the first input NMOS FET **202** of the output switching block **504** to operate in its on-state. Therefore, during dead times, current may flow from switching node **110** to the respective output power node **112** of the output switching block **504** via the body diode **218** of the output NMOS FET **204** of the output switching block **504**. Additionally, control signal **rc** of the output switching block **504** is logic high in the second off-state **1006** to cause the reference NMOS FET **512** to operate in its on-state. Therefore, the output switching block **504** provides a path for current $I_{sub.L}$ to flow through the reference NMOS FET **512** of the output switching block **504** in the second off-state **1006**. Control signal **bc** of the output switching block **504** is also logic high in the second off-state **1006** to cause the PMOS transistor **524** of the output switching block **504** to operate in its on-state. Consequently, the bootstrap capacitor **522** of the output switching block **504** charges from the auxiliary power supply rail **126** and through the PMOS FET **524** of the output switching block **504** in the second off-state **1006** of the output switching block **504**. The controller **106** causes the output switching block **504** to operate in the second off-state **1006** instead of the first off-state **1004**, for example, if the output switching block **504** is expected to quickly transition back to the on-state **1002**, such as when the output NMOS FET **204** of the output switching block **504** needs to switch between its on-state and its off-state multiple times without the output NMOS FETs **204** of other output switching blocks **504** needing to operate in their respective on-states in the meantime. Operation in the second off-state **1006** instead of in the first off-state **1004** is this scenario advantageously eliminates losses caused by unnecessary switching of the input NMOS FET **202**.

[0042] FIG. **11** includes graphs **1100**, **1102**, **1104**, **1106**, and **1108** collectively illustrating an example transition **1008** (FIG. **10**) of an output switching block **504** from the first off-state **1004** to the on-state **1002**. Graph **1100** is of the control signal **pc3** versus time, graph **1102** is of the control signal **sbin** versus time, graph **1104** is of the control signal **sbout** versus time, graph **1106** is of the control signal **rc** versus time, and graph **1108** is of the control signal **bc** versus time. Graphs **1100**, **1102**, **1104**, **1106**, and **1108** have a common time base. The output switching block **504** transitions from the first off-state **1004** to the on-state **1002** at a time $t_{sub.1}$ in the FIG. **11** example. The output switching block **504** accordingly operates in the first off-state **1004** before the time $t_{sub.1}$, wherein each of the control signal **sbin** and the control signal **sbout** is logic low, and each of the control signal **rc** and the control signal **bc** is logic high. Additionally, the control signal **pc3** is logic high during the first off-state **1004** in the FIG. **11** example. The output switching block **504** transitions from the first off-state **1004** to the on-state **1002** at the time $t_{sub.1}$, and each of the control signal **rc** and the control signal **bc** transitions from logic high to logic low at time $t_{sub.1}$. The control signal **sbin** next transitions from logic low to logic high, followed by the control signal **pc3** transitioning from logic high to logic low. Finally, the control signal **sbout** transitions from logic low to logic high after the control signal **pc3** has transitioned from logic high to logic low. While power conversion switching device **120** can no longer provide a path for the current $I_{sub.L}$ after the control signal **pc3** transitions to logic low in the on-state **1002**, the output switching block **504** provides a path for the current $I_{sub.L}$ to a load powered from its respective output power node **112** during the on-state **1002**.

[0043] FIG. **12** includes graphs **1200**, **1202**, **1204**, **1206**, and **1208** collectively illustrating an example transition **1010** (FIG. **10**) of an output switching block **504** from the on-state **1002** to the first off-state **1004**. Graph **1200** is of the control signal **pc3** versus time, graph **1202** is of the control signal **sbin** versus time, graph **1204** is of the control signal **sbout** versus time, graph **1206** is of the control signal **rc** versus time, and graph **1208** is of the control signal **bc** versus time. Graphs **1200**, **1202**, **1204**, **1206**, and **1208** have a common time base. The output switching block **504** transitions from the on-state **1002** to the first off-state **1004** at a time $t_{sub.2}$ in the FIG. **12** example. The output switching block **504** accordingly operates in the on-state **1002** before the time $t_{sub.2}$, wherein each of the control signal **sbin** and the control signal **sbout** is logic high, and each of the control signal **rc** and the control signal **bc** is logic low. Additionally, the control signal **pc3** is

logic low during the on-state **1002** in the FIG. **12** example. The output switching block **504** transitions from the on-state **1002** to the first off-state **1004** at the time $t_{sub.2}$, and the control signal **pc3** accordingly transitions from logic high to logic low, followed by the control signal **pc3** transitioning from logic low to logic high. The control signal **sbin** transitions from logic high to logic low after control signal **pc3** transitions to logic high. Finally, the control signal **rc** and the control signal **bc** transition from logic low to logic high together after the control signal **sbin** has transitioned from logic high to logic low. Once the power conversion switching device **120** is in its on-state, the output switching block **504** no longer needs to provide a path for the current $I_{sub.L}$. [0044] FIG. **13** includes graphs **1300**, **1302**, **1304**, **1306**, and **1308** collectively illustrating an example transition **1012** (FIG. **10**) of an output switching block **504** from the second off-state **1006** to the on-state **1002**. Graph **1300** is of the control signal **pc3** versus time, graph **1302** is of the control signal **sbin** versus time, graph **1304** is of the control signal **sbout** versus time, graph **1306** is of the control signal **rc** versus time, and graph **1308** is of the control signal **bc** versus time. Graphs **1300**, **1302**, **1304**, **1306**, and **1308** have a common time base. The output switching block **504** transitions from the second off-state **1006** to the on-state **1002** at a time $t_{sub.3}$ in the FIG. **13** example. The output switching block **504** accordingly operates in the second off-state **1006** before the time $t_{sub.3}$, wherein the control signal **sbin** is logic high, the control signal **sbout** is logic low, and each of the control signal **rc** and the control signal **bc** is logic high. Additionally, the control signal **pc3** is logic high during the second off-state **1006** in the FIG. **13** example. The output switching block **504** transitions from the second off-state **1006** to the on-state **1002** at the time $t_{sub.3}$, and the control signal **rc** and the control signal **bc** transition from logic high to logic low together. Next, the control signal **pc3** transitions from logic high to logic low after each of the control signal **rc** and the control signal **bc** have transitioned from logic high to logic low, and the control signal **sbout** transitions from logic low to logic high after the control signal **pc3** has transitioned to logic low. The output switching block **504** provides a path for the current $I_{sub.L}$ to a load powered from its respective output power node **112** once the control signal **sbout** is logic high.

[0045] FIG. **14** includes graphs **1400**, **1402**, **1404**, **1406**, and **1408** collectively illustrating an example transition **1014** (FIG. **10**) of an output switching block **504** from the on-state **1002** to the second off-state **1006**. Graph **1400** is of the control signal **pc3** versus time, graph **1402** is of the control signal **sbin** versus time, graph **1404** is of the control signal **sbout** versus time, graph **1406** is of the control signal **rc** versus time, and graph **1408** is of the control signal **bc** versus time. Graphs **1400**, **1402**, **1404**, **1406**, and **1408** have a common time base. The output switching block **504** transitions from the on-state **1002** to the second off-state **1006** at a time $t_{sub.4}$ in the FIG. **14** example. The output switching block **504** accordingly operates in the on-state **1002** before the time $t_{sub.4}$, wherein each of the control signal **sbin** and the control signal **sbout** is logic high, and each of the control signal **rc** and the control signal **bc** is logic low. Additionally, the control signal **pc3** is logic low during the on-state **1002** in the FIG. **14** example. The output switching block **504** transitions from the on-state **1002** to the second off-state **1006** at the time $t_{sub.4}$, and the control signal **sbout** accordingly transitions from logic high to logic low. The control signal **pc3** transitions from logic low to logic high after the control signal **sbout** is logic low. Finally, the control signal **rc** and the control signal **bc** transition from logic low to logic high together after the control signal **pc3** is logic high. Consequently, there is a path for the current $I_{sub.L}$ through both of the power conversion switching device **120** and the reference NMOS FET **512** of the output switching block **504** in the second off-state **1006** after switching transitions are complete slightly after time $t_{sub.4}$.

[0046] Referring again to FIG. 5-7, any of the output switching blocks **504** could be modified to include one or more additional elements. For example, FIG. **15** is a schematic diagram of an output switching block **1504(1)**, which is an alternate embodiment of the output switching block **504(1)** (FIG. **5**) further including elements to perform current sensing. It is understood that the other output switching blocks **504** could be modified in a manner similar to that illustrated in FIG. **15**.

The output switching block **1504(1)** differs from the output switching block **504(1)** in that output switching block **1504(1)** further includes first replica circuitry **1528**, second replica circuitry **1530**, and a current sense comparator **1532**. Each of the first replica circuitry **1528** and the second replica circuitry **1530** is a scaled model of the output NMOS FET **204(1)**. Additionally, each of the first replica circuitry **1528** and the second replica circuitry **1530** operates in the same power supply domain as the switching device driver circuitry of the output switching block **1504(1)**, i.e., between the bootstrap power supply rail **220(1)** and the internal node **214(1)**. The first replica circuitry **1528** generates a signal **rs1** in response to voltage at the internal node **214(1)**, and the second replica circuitry **1530** generates a signal **rs2** in response to voltage at the output power node **112(1)**. The current sense comparator **1532** compares the signal **rs1** and the signal **rs2** to generate a signal **i_s** representing magnitude of current flowing through the output NMOS FET **204(1)**.

Combinations of Features

[0047] Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations.

[0048] (A1) A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes a power conversion stage and a plurality of output switching blocks. The power conversion stage is electrically coupled between an input power node and a switching node, and the power conversion stage includes a first power conversion switching device electrically coupled between the switching node and a reference node. Each output switching block is electrically coupled between the switching node and a respective output power node. Each output switching block includes (1) a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block that is electrically coupled between the switching node and a respective internal node of the output switching block, where the input NMOS FET includes a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block, and (2) a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block that is electrically coupled between the respective internal node of the output switching block and the respective output power node of the output switching block, where the output NMOS FET includes a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the respective output power node of the output switching block and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block.

[0049] (A2) In the SIMO DC-to-DC converter denoted as (A1), each output switching block may further include (1) respective bootstrap circuitry of the output switching block that is configured to generate a respective bootstrap power supply rail of the output switching block referenced to the respective internal node of the output switching block, (2) respective input switch driver circuitry of the output switching block that is configured to drive a gate of the respective input NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block, and (3) respective output switch driver circuitry of the output switching block that is configured to drive a gate of the respective output NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block.

[0050] (A3) The SIMO DC-to-DC converter denoted as (A1) may further include a controller configured to control at least each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block.

[0051] (A4) In the SIMO DC-to-DC converter denoted as (A3), the controller may be further configured to control each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output

switching block in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

[0052] (A5) In either one of the SIMO DC-to-DC converters denoted as (A3) or (A4), the controller may be further configured to control the first power conversion switching device such that the first power conversion switching device operates in its on-state solely when the respective output NMOS FET of each output switching block operates in its respective off-state.

[0053] (A6) In any one of the SIMO DC-to-DC converters denoted as (A1) through (A5), (i) each output switching block may further include a respective reference switching device electrically coupled between the respective internal node of the output switching block and the reference node, (ii) each output switching block may further include a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series between the respective internal node of the output switching block and an auxiliary power supply rail.

[0054] (A7) In the SIMO DC-to-DC converter denoted as (A6), the controller may be further configured, with respect to each output switching block, to control the first power conversion switching device to cause the first power conversion switching device to switch from its on-state to its off-state after each of (i) the respective reference switching device of the output switching block has switched from its on-state to its off-state and (ii) the respective bootstrap control switching device of the output switching block has switched from its on-state to its off-state.

[0055] (A8) In either one of the SIMO DC-to-DC converters denoted as (A6) or (A7), the controller may be further configured to control the respective reference switching device of each output switching block and the respective bootstrap control switching device of each output switching block such that in a given output switching block, the respective reference switching device of the output switching block and the respective bootstrap control switching device of the output switching block do not switch from their respective off-states to their respective on-state until after (i) the respective output NMOS FET of the output switching block has switched from its on-state to its off-state and (ii) the first power conversion switching device has switched from its off-state to its on-state.

[0056] (A9) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A8), wherein the controller may be further configured to control the respective output NMOS FET of each output switching block in a manner such that at most one output NMOS FET is in its on-state at any given time during operation of the SIMO DC-to-DC converter.

[0057] (A10) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A9), the controller may be further configured to control the respective output NMOS FET of each output switching block such that each output NMOS FET does not switch from its off-state to its on-state until after the first power conversion device switches from its on-state to its off-state.

[0058] (A11) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A10), the controller may be further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its off-state to its on-state before the first power conversion switching device switches from its on-state to its off-state.

[0059] (A12) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A11), the controller may be further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its on-state to its off-state after the first power conversion switching device switches from its off-state to its on-state.

[0060] (A13) In the SIMO DC-to-DC converter denoted as (A2), the respective bootstrap circuitry of each output switching block may include (i) a respective bootstrap capacitor electrically coupled between the respective bootstrap power supply rail of the output switching block and the respective internal node of the output switching block and (ii) a respective bootstrap control switching device electrically coupled between an auxiliary power supply rail and the respective bootstrap capacitor

of the output switching block.

[0061] (A14) In any one of the SIMO DC-to-DC converters denoted as (A1) through (A13), the power conversion stage may have a buck and boost topology.

[0062] (B1) A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes a power conversion stage, a first output switching block, and a second output switching block. The power conversion stage electrically is coupled between an input power node and a switching node, and the power conversion stage includes a first power conversion switching device electrically coupled between the switching node and a reference node. The first output switching block includes (1) a first input enhancement mode, N-type metal oxide semiconductor field effect transistor (first input NMOS FET) electrically coupled between the switching node and a first internal node, where the first input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the first internal node, and (2) a first output enhancement mode, N-type metal oxide semiconductor field effect transistor (first output NMOS FET) electrically coupled between the first internal node and a first output power node, where the first output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the first output power node and (ii) an anode of the body diode is electrically coupled to the first internal node. The second output switching block includes (1) a second input enhancement mode, N-type metal oxide semiconductor field effect transistor (second input NMOS FET) electrically coupled between the switching node and a second internal node, where the second input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the second internal node, and (2) a second output enhancement mode, N-type metal oxide semiconductor field effect transistor (second output NMOS FET) electrically coupled between the second internal node and a second output power node, where the second output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the second output power node and (ii) an anode of the body diode is electrically coupled to the second internal node.

[0063] (B2) In the SIMO DC-to-DC converter denoted as (B1), (1) the first output switching block may further include (i) first bootstrap circuitry configured to generate a first bootstrap power supply rail referenced to the first internal node, (ii) first input switch driver circuitry configured to drive a gate of the first input NMOS FET and being electrically powered by the first bootstrap power supply rail, and (iii) first output switch driver circuitry configured to drive a gate of the first output NMOS FET and being electrically powered by the first bootstrap power supply rail, and (2) the second output switching block may further include (i) second bootstrap circuitry configured to generate a second bootstrap power supply rail referenced to the second internal node, (ii) second input switch driver circuitry configured to drive a gate of the second input NMOS FET and being electrically powered by the second bootstrap power supply rail, and (iii) second output switch driver circuitry configured to drive a gate of the second output NMOS FET and being electrically powered by the second bootstrap power supply rail.

[0064] (B3) Either one of the SIMO DC-to-DC converters denoted as (B1) and (B2) may further include a controller configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET.

[0065] (B4) In the SIMO DC-to-DC converter denoted as (B3), the controller may be further configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

[0066] (B5) Any one of the SIMO DC-to-DC converters denoted as (B1) through (B4) may further

include a third output switching block including (1) a third input enhancement mode, N-type metal oxide semiconductor field effect transistor (third input NMOS FET) electrically coupled between the switching node and a third internal node, where the third input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the third internal node, and (2) a third output enhancement mode, N-type metal oxide semiconductor field effect transistor (third output NMOS FET) electrically coupled between the third internal node and a third output power node, where the third output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the third output power node and (ii) an anode of the body diode is electrically coupled to the third internal node.

[0067] (C1) A method for controlling a plurality of output switching blocks in a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes, in each of the plurality of output switching blocks, (1) generating a respective bootstrap power supply rail of the output switching block referenced to a respective internal node of the output switching block, (2) driving a gate of a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, where a drain of the respective input NMOS FET of the output switching block is electrically coupled to a switching node of the SIMO DC-to-DC converter, and a source of the respective input NMOS FET of the output switching block is electrically coupled to the respective internal node of the output switching block, and (3) driving a gate of a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, where a drain of the respective output NMOS FET of the output switching block is electrically coupled to a respective output power node of the output switching block, and a source of the respective output NMOS FET of the output switching block is electrically coupled to the respective internal node of the output switching block.

[0068] (C2) The method denoted as (C1) may further include controlling the respective input NMOS FET of each output switching block and the respective output NMOS FET of each output switching block such that the respective output power node of at most one output switching block at any given time is powered by the SIMO DC-to-DC converter during operation of the SIMO DC-to-DC converter.

[0069] Changes may be made in the above methods, devices, and systems without departing from the scope hereof. For example, power conversion stage **102** could be modified to have a different topology, such a buck topology or a boost topology instead of a buck and boost topology. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

Claims

1. A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, comprising: a power conversion stage electrically coupled between an input power node and a switching node, the power conversion stage including a first power conversion switching device electrically coupled between the switching node and a reference node; and a plurality of output switching blocks, each output switching block being electrically coupled between the switching node and a respective output power node, each output switching block including: a respective input

enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block that is electrically coupled between the switching node and a respective internal node of the output switching block, the input NMOS FET including a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block, and a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block that is electrically coupled between the respective internal node of the output switching block and the respective output power node of the output switching block, the output NMOS FET including a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the respective output power node of the output switching block and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block.

2. The SIMO DC-to-DC converter of claim 1, wherein each output switching block further includes: respective bootstrap circuitry of the output switching block that is configured to generate a respective bootstrap power supply rail of the output switching block referenced to the respective internal node of the output switching block, respective input switch driver circuitry of the output switching block that is configured to drive a gate of the respective input NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block, and respective output switch driver circuitry of the output switching block that is configured to drive a gate of the respective output NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block.

3. The SIMO DC-to-DC converter of claim 1, further comprising a controller configured to control at least each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block.

4. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

5. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the first power conversion switching device such that the first power conversion switching device operates in its on-state solely when the respective output NMOS FET of each output switching block operates in its respective off-state.

6. The SIMO DC-to-DC converter of claim 3, wherein: each output switching block further includes a respective reference switching device electrically coupled between the respective internal node of the output switching block and the reference node; each output switching block further includes a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series between the respective internal node of the output switching block and an auxiliary power supply rail; and with respect to each output switching block, the controller is further configured to control the first power conversion switching device to cause the first power conversion switching device to switch from its on-state to its off-state after each of (i) the respective reference switching device of the output switching block has switched from its on-state to its off-state and (ii) the respective bootstrap control switching device of the output switching block has switched from its on-state to its off-state.

7. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective output NMOS FET of each output switching block in a manner such that at most one output NMOS FET is in its on-state at any given time during operation of the SIMO DC-to-DC converter.

8. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to

control the respective output NMOS FET of each output switching block such that each output NMOS FET does not switch from its off-state to its on-state until after the first power conversion device switches from its on-state to its off-state.

9. The SIMO DC-to-DC converter of claim 3, wherein: each output switching block further includes a respective reference switching device electrically coupled between the respective internal node of the output switching block and the reference node; each output switching block further includes a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series between the respective internal node of the output switching block and an auxiliary power supply rail; and the controller is further configured to control the respective reference switching device of each output switching block and the respective bootstrap control switching device of each output switching block such that in a given output switching block, the respective reference switching device of the output switching block and the respective bootstrap control switching device of the output switching block do not switch from their respective off-states to their respective on-state until after (i) the respective output NMOS FET of the output switching block has switched from its on-state to its off-state and (ii) the first power conversion switching device has switched from its off-state to its on-state.

10. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its off-state to its on-state before the first power conversion switching device switches from its on-state to its off-state.

11. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its on-state to its off-state after the first power conversion switching device switches from its off-state to its on-state.

12. The SIMO DC-to-DC converter of claim 2, wherein the respective bootstrap circuitry of each output switching block includes: a respective bootstrap capacitor electrically coupled between the respective bootstrap power supply rail of the output switching block and the respective internal node of the output switching block; and a respective bootstrap control switching device electrically coupled between an auxiliary power supply rail and the respective bootstrap capacitor of the output switching block.

13. The SIMO DC-to-DC converter of claim 1, wherein the power conversion stage has a buck and boost topology.

14. A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, comprising: a power conversion stage electrically coupled between an input power node and a switching node, the power conversion stage including a first power conversion switching device electrically coupled between the switching node and a reference node; a first output switching block including: a first input enhancement mode, N-type metal oxide semiconductor field effect transistor (first input NMOS FET) electrically coupled between the switching node and a first internal node, the first input NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the first internal node, and a first output enhancement mode, N-type metal oxide semiconductor field effect transistor (first output NMOS FET) electrically coupled between the first internal node and a first output power node, the first output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the first output power node and (ii) an anode of the body diode is electrically coupled to the first internal node; and a second output switching block including: a second input enhancement mode, N-type metal oxide semiconductor field effect transistor (second input NMOS FET) electrically coupled between the switching node and a second internal node, the second input NMOS FET including a body diode

oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the second internal node, and a second output enhancement mode, N-type metal oxide semiconductor field effect transistor (second output NMOS FET) electrically coupled between the second internal node and a second output power node, the second output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the second output power node and (ii) an anode of the body diode is electrically coupled to the second internal node.

15. The SIMO DC-to-DC converter of claim 14, wherein: the first output switching block further includes: first bootstrap circuitry configured to generate a first bootstrap power supply rail referenced to the first internal node, first input switch driver circuitry configured to drive a gate of the first input NMOS FET and being electrically powered by the first bootstrap power supply rail, and first output switch driver circuitry configured to drive a gate of the first output NMOS FET and being electrically powered by the first bootstrap power supply rail; and the second output switching block further includes: second bootstrap circuitry configured to generate a second bootstrap power supply rail referenced to the second internal node, second input switch driver circuitry configured to drive a gate of the second input NMOS FET and being electrically powered by the second bootstrap power supply rail, and second output switch driver circuitry configured to drive a gate of the second output NMOS FET and being electrically powered by the second bootstrap power supply rail.

16. The SIMO DC-to-DC converter of claim 14, further comprising a controller configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET.

17. The SIMO DC-to-DC converter of claim 14, wherein the controller is further configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

18. The SIMO DC-to-DC converter of claim 14, further comprising a third output switching block including: a third input enhancement mode, N-type metal oxide semiconductor field effect transistor (third input NMOS FET) electrically coupled between the switching node and a third internal node, the third input NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the third internal node; and a third output enhancement mode, N-type metal oxide semiconductor field effect transistor (third output NMOS FET) electrically coupled between the third internal node and a third output power node, the third output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the third output power node and (ii) an anode of the body diode is electrically coupled to the third internal node.

19. A method for controlling a plurality of output switching blocks in a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, the method comprising, in each of the plurality of output switching blocks: generating a respective bootstrap power supply rail of the output switching block referenced to a respective internal node of the output switching block; driving a gate of a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, a drain of the respective input NMOS FET of the output switching block being electrically coupled to a switching node of the SIMO DC-to-DC converter, and a source of the respective input NMOS FET of the output switching block being electrically coupled to the respective internal node of the output switching block; and driving a gate of a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block between a voltage of the respective bootstrap

power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, a drain of the respective output NMOS FET of the output switching block being electrically coupled to a respective output power node of the output switching block, and a source of the respective output NMOS FET of the output switching block being electrically coupled to the respective internal node of the output switching block.

20. The method of claim 19, further comprising controlling the respective input NMOS FET of each output switching block and the respective output NMOS FET of each output switching block such that the respective output power node of at most one output switching block at any given time is powered by the SIMO DC-to-DC converter during operation of the SIMO DC-to-DC converter.
